```
num_queries = 20

for i in range(num_queries):
    print("iteration", i)
    print("incumbent", train_x[train_y.argmax()], train_y.max())

    model, likelihood = fit_gp_model(train_x, train_y)

    policy = ... ❶

    next_x, acq_val = botorch.optim.optimize_acqf(
        policy,
        bounds=bounds,
        q=1,
        num_restarts=40, ❷
        raw_samples=100, ❷
    )

    visualize_progress_and_policy(policy, next_x=next_x) ❸

    next_y = f(next_x)

    train_x = torch.cat([train_x, next_x])
    train_y = torch.cat([train_y, next_y])
```

❶ Placeholder for policy initialization.

❷ Making the search more exhaustive.

❸ Calling the new visualization helper function.

6. Run the Probability of Improvement (PoI) policy on this objective function. Observe that the policy once again gets stuck at a local optimum. Replace the line that initializes the BayesOpt policy with:

```
policy = botorch.acquisition.analytic.ProbabilityOfImprovement(
    model, best_f=train_y.max()
)
```

Running the entire notebook shows that the policy once again gets stuck at a local optimum, as shown in Figure A.6.

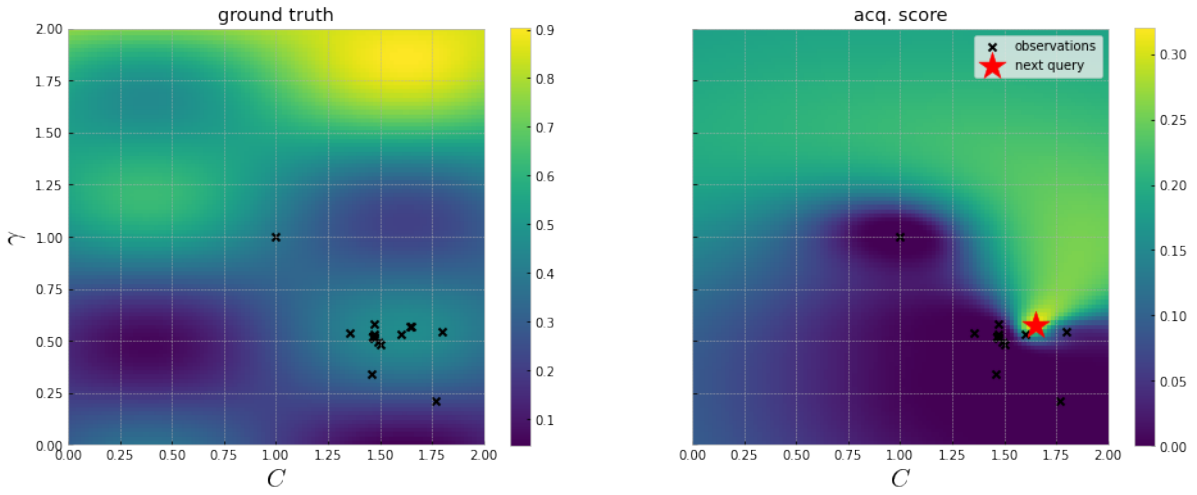


Figure A.6 Optimization progress of Probability of Improvement at the last iteration. The policy is stuck at a local optimum in the lower right corner.

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